

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Winter Examination – 2023

Course: B. Tech Branch: Electrical Engineering Semester :IV

Subject Code & Name: BTEEE406A_Y19 Solid State Devices

Max Marks: 60

Date:29/01/2024

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q.1 Solve Any Two of the following.		12
A) Describe the applications of zener diodes in electronic circuits. How does a zener diode regulate voltage?	CO1	6
B) Elaborate working of BJT with a neat schematic and explain in brief CB, CE, CC configurations of BJT.	CO1	6
C) What is MOSFET biasing? Differentiate NMOS, PMOS and CMOS.	CO1	6
Q.2 Solve Any Two of the following.		12
A) What are signal and power amplifiers? Differentiate them based on characteristics and application.	CO4	6
B) Compare the low-frequency response characteristics of FET amplifiers with those of transistor amplifiers. What similarities and differences exist in the techniques used to enhance low-frequency performance in these devices?	CO5	6
C) What is the role of feedback in signal amplifiers? How does feedback influence the gain, bandwidth, and distortion of an amplifier?	CO5	6
Q.3 Solve Any Two of the following.		12
A) Define an operational amplifier (Op-Amp). What are its key characteristics that make it suitable for various applications?	CO2 & CO4	6
B) Draw the equivalent circuit of an operational amplifier (op-amp) and label its key components. Explain how closed loop configurations affect the overall behavior of the op-amp circuit.	CO2 & CO4	6
C) Draw ideal voltage transfer curve and explain in brief its features.	CO4	6
Q.4 Solve Any Two of the following.		12
A) Differentiate active low pass filter and active high pass filter.	CO1	6
B) Write short note on following: a) Comparators b) Oscillators	CO1	6
C) Explain the basic principle of operation of a voltage to current converter. Include the key components and their roles in the conversion process.	CO4	6
Q.5 Solve Any Two of the following.		12
A) Differentiate DC amplifiers and AC amplifiers.	CO4	6
B) Describe the operation of the 555 timer in astable mode. Include the role of	CO3	6

- external components and the resulting waveform.
- C) What is voltage regulator? Explain in brief fixed and adjustable voltage regulator.

CO3

6

***** End *****